The name of the exam 22 gennaio 2022	Prof.	Firma leggibile dello studente
Cognome:	Nome:	Matricola:

INSTRUCTIONS

- Write here your instructions
- two
- three

Part One

15 punti

SOME EXERCISES FROM 2 FILES

1. *(1 punto)* exercise 2a $a = 4$, $b = 2$, $c = 7$

(a) answer 5 wrong

(b) answer 4 wrong

(c) answer 2 wrong

(d) answer 3 wrong

► (e) answer 1 correct

2. *(2 punti)* exercise 3b $a = 2$, $b = 3$, $c = 6$

(a) answer 2 wrong

(b) answer 4 wrong

► (c) answer 1 correct

(d) answer 5 wrong

(e) answer 3 wrong

Soluzione: answer 1

3. *(1 punto)* exercise 2b $a = 3$, $b = 3$, $c = 5$

(a) answer 2 wrong

(b) answer 4 wrong

(c) answer 5 wrong

(d) answer 3 wrong

► (e) answer 1 correct

4. *(1 punto)* exercise 2c $a = 3$, $b = 5$, $c = 6$

► (a) answer 1 correct

(b) answer 4 wrong

(c) answer 2 wrong

(d) answer 5 wrong

(e) answer 3 wrong

5. *(2 punti)* exercise 3a $a = 5$, $b = 3$, $c = 3$

► (a) answer 1 correct

(b) answer 5 wrong

(c) answer 4 wrong

(d) answer 2 wrong

(e) answer 3 wrong

Soluzione: answer 1

6. *(1 punto)* exercise 1a $a = 7$

(a) answer 3 wrong

(b) answer 2 wrong

► (c) answer 1 correct

7. *(2 punti)* exercise 9b $a = 3$, $b = 5$, $c = 3$

(a) answer 3 wrong

► (b) answer 1 correct

(c) answer 2 wrong

(d) answer 4 wrong

(e) answer 5 wrong

8. *(1 punto)* exercise 14a $a = 3$, $b = 5$, $c = 5$ $alphai = 11$ $alphaii = 5$

(a) answer 5 wrong

(b) answer 3 wrong

(c) answer 2 wrong

► (d) answer 1 correct

(e) answer 4 wrong

Soluzione: This is the solution $a = 3$, $b = 5$, $c = 5$

9. *(1 punto)* exercise 4e $a = 2$, $b = 5$, $c = 7$

(a) answer 2 wrong

(b) answer 4 wrong

(c) answer 5 wrong

(d) answer 3 wrong

► (e) answer 1 correct

10. *(1 punto)* $a = 15$, $b = 15$, $c = 3$, $k = 5$
Evaluate with two exact decimals

$15 : 5 =$ **3**

All the problems of a file

11. *(1 punto)* exercise 11a

$$\{b,\ c,\ y\}$$

► (a) answer 1 correct

(b) answer 3 wrong

(c) answer 2 wrong

12. *(1 punto)* exercise 11b

$$\{a,\ b,\ c\}$$

(a) answer 3 wrong

► (b) answer 1 correct

(c) answer 2 wrong

Some other questions

1. *(1 punto)* exercise 12a $a = 3$, $b = 5$, $c = 6$

► (a) answer 1 correct

(b) answer 4 wrong

(c) answer 3 wrong

(d) answer 2 wrong

(e) answer 5 wrong

2. *(2 punti)* exercise 13b $a = 4$, $b = 4$, $c = 6$

(a) answer 2 wrong

(b) answer 3 wrong

(c) answer 4 wrong

(d) answer 5 wrong

► (e) answer 1 correct

Part two

Some other instructions.

ESERCIZIO 1. Let $A = \{a,b,c\}$ and $B = \{a,c,x\}$.

4 punti

(a) *(2 punti)* List (without repetition) the elements of the set $A \cup B$

Soluzione:

$$A \cup B = \{a,b,c,x\}$$

(b) *(2 punti)* List (without repetition) the elements of the set $A \cap B$

Soluzione:

$$A \cap B = \{a,c\}$$

An example of matching question.

Esercizio 2.
$a = 5$, $b = 4$, $c = 6$

(A)	5×4	(A)	20
(C)	$5 + 4$	(B)	1
(B)	$6 - 5$	(C)	9

3 punti

An example of “tabella” question.

Esercizio 3. Solve the following equations:

4 punti	

Equation	Solution
$x^2 + 7x + 10 = 0$	$x = 5;\ x = 2$
$x^2 - 7x + 10 = 0$	$x = -5;\ x = -2$
$x^2 + 3x - 10 = 0$	$x = -5;\ x = 2$
$x^2 - 3x - 10 = 0$	$x = 5;\ x = -2$

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Part One

15 punti

SOME EXERCISES FROM 2 FILES

1. *(1 punto)* exercise 2c $a = 2$, $b = 2$, $c = 5$

(a) answer 5 wrong

(b) answer 4 wrong

(c) answer 3 wrong

► (d) answer 1 correct

(e) answer 2 wrong

2. *(2 punti)* exercise 3a $a = 2$, $b = 2$, $c = 3$

► (a) answer 1 correct

(b) answer 4 wrong

(c) answer 5 wrong

(d) answer 2 wrong

(e) answer 3 wrong

Soluzione: answer 1

3. *(1 punto)* exercise 2b $a = 4$, $b = 3$, $c = 8$

(a) answer 2 wrong

(b) answer 3 wrong

► (c) answer 1 correct

(d) answer 5 wrong

(e) answer 4 wrong

4. *(2 punti)* exercise 3d $a = 5$, $b = 4$, $c = 4$

(a) answer 2 wrong

► (b) answer 1 correct

(c) answer 5 wrong

(d) answer 4 wrong

(e) answer 3 wrong

Soluzione: answer 1

5. *(1 punto)* exercise 2a $a = 3$, $b = 3$, $c = 3$

(a) answer 3 wrong

(b) answer 2 wrong

(c) answer 5 wrong

► (d) answer 1 correct

(e) answer 4 wrong

6. *(1 punto)* exercise 1b $a = 2$

► (a) answer 1 correct

(b) answer 3 wrong

(c) answer 2 wrong

7. *(1 punto)* $a = 14$, $b = 14$, $c = 4$, $k = 5$
Evaluate with two exact decimals

$14 : 4 =$ **3,5**

8. *(2 punti)* exercise 9c $a = 5$, $b = 3$, $c = 8$

► (a) answer 1 correct

(b) answer 5 wrong

(c) answer 4 wrong

(d) answer 3 wrong

(e) answer 2 wrong

9. *(1 punto)* exercise 14a $a = 3$, $b = 3$, $c = 2$ $alphai = 5$ $alphaii = 7$

(a) answer 3 wrong

► (b) answer 1 correct

(c) answer 4 wrong

(d) answer 5 wrong

(e) answer 2 wrong

Soluzione: This is the solution $a = 3$, $b = 3$, $c = 2$

10. *(1 punto)* exercise 4b $a = 5$, $b = 5$, $c = 2$

(a) answer 5 wrong

► (b) answer 1 correct

(c) answer 3 wrong

(d) answer 2 wrong

(e) answer 4 wrong

All the problems of a file

11. *(1 punto)* exercise 11b

$$\{c,\ y,\ z\}$$

(a) answer 3 wrong

(b) answer 2 wrong

► (c) answer 1 correct

12. *(1 punto)* exercise 11a

$$\{a,\ c,\ x\}$$

(a) answer 3 wrong

(b) answer 2 wrong

► (c) answer 1 correct

Some other questions

1. *(2 punti)* exercise 13c $a = 5$, $b = 5$, $c = 8$

► (a) answer 1 correct

(b) answer 2 wrong

(c) answer 3 wrong

(d) answer 4 wrong

(e) answer 5 wrong

2. *(1 punto)* exercise 12c $a = 4$, $b = 2$, $c = 5$

(a) answer 2 wrong

(b) answer 5 wrong

(c) answer 3 wrong

► (d) answer 1 correct

(e) answer 4 wrong

Part two

Some other instructions.

ESERCIZIO 1. Let $A = \{a,b,c\}$ and $B = \{a,c,y\}$.

4 punti

(a) *(2 punti)* List (without repetition) the elements of the set $A \cup B$

Soluzione:

$$A \cup B = \{a,b,c,y\}$$

(b) *(2 punti)* List (without repetition) the elements of the set $A \cap B$

Soluzione:

$$A \cap B = \{a,c\}$$

An example of matching question.

Esercizio 2. $a = 12$, $b = 14$, $c = 4$

(B)	12×14	(A)	168
(A)	12×14	(B)	168
(C)	12×14	(C)	168
(E)	$48 : 12$	(D)	256
(D)	4^4	(E)	4

3 punti

An example of “tabella” question.

Esercizio 3. Complete the following table of derivatives:

5 punti	

Function	Derivative
$f(x) = x^2$	$f'(x) = 2x$
$f(x) = \sin x$	$f'(x) = \cos x$
$f(x) = \cos x$	$f'(x) = -\sin x$

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Part One

SOME EXERCISES FROM 2 FILES

1. *(1 punto)* exercise 2c $a = 5$, $b = 5$, $c = 2$

► (a) answer 1 correct

(b) answer 2 wrong

(c) answer 3 wrong

(d) answer 4 wrong

(e) answer 5 wrong

2. *(2 punti)* exercise 3a $a = 5$, $b = 4$, $c = 4$

► (a) answer 1 correct

(b) answer 4 wrong

(c) answer 3 wrong

(d) answer 2 wrong

(e) answer 5 wrong

Soluzione: answer 1

3. *(1 punto)* exercise 2a $a = 2$, $b = 5$, $c = 5$

► (a) answer 1 correct

(b) answer 2 wrong

(c) answer 4 wrong

(d) answer 3 wrong

(e) answer 5 wrong

4. *(2 punti)* exercise 3b $a = 5$, $b = 5$, $c = 8$

(a) answer 4 wrong

(b) answer 5 wrong

► (c) answer 1 correct

(d) answer 3 wrong

(e) answer 2 wrong

Soluzione: answer 1

15 punti

5. *(1 punto)* exercise 2b $a = 5$, $b = 2$, $c = 6$

(a) answer 4 wrong

(b) answer 3 wrong

► (c) answer 1 correct

(d) answer 2 wrong

(e) answer 5 wrong

6. *(1 punto)* exercise 14a $a = 4$, $b = 3$, $c = 7$ $alphai = 3$ $alphaii = 11$

(a) answer 3 wrong

► (b) answer 1 correct

(c) answer 5 wrong

(d) answer 2 wrong

(e) answer 4 wrong

Soluzione: This is the solution $a = 4$, $b = 3$, $c = 7$

7. *(1 punto)* $a = 13$, $b = 13$, $c = 3$, $k = 4$
Evaluate with two exact decimals

$13 : 4 =$ **3,25**

8. *(1 punto)* exercise 4c $a = 3$, $b = 2$, $c = 8$

► (a) answer 1 correct

(b) answer 3 wrong

(c) answer 2 wrong

(d) answer 4 wrong

(e) answer 5 wrong

9. *(1 punto)* exercise 1a $a = 3$

(a) answer 3 wrong

(b) answer 2 wrong

► (c) answer 1 correct

10. *(2 punti)* exercise 9d $a = 3$, $b = 3$, $c = 5$

► (a) answer 1 correct

(b) answer 2 wrong

(c) answer 5 wrong

(d) answer 4 wrong

(e) answer 3 wrong

All the problems of a file

11. *(1 punto)* exercise 11b

$$\{a,\ b, c\}$$

(a) answer 3 wrong

► (b) answer 1 correct

(c) answer 2 wrong

12. *(1 punto)* exercise 11a

$$\{a,\ b, x\}$$

► (a) answer 1 correct

(b) answer 2 wrong

(c) answer 3 wrong

Some other questions

1. *(1 punto)* exercise 12a $a = 3$, $b = 5$, $c = 3$

(a) answer 5 wrong

► (b) answer 1 correct

(c) answer 4 wrong

(d) answer 3 wrong

(e) answer 2 wrong

2. *(2 punti)* exercise 13d $a = 3$, $b = 4$, $c = 6$

(a) answer 3 wrong

(b) answer 2 wrong

(c) answer 4 wrong

► (d) answer 1 correct

(e) answer 5 wrong

Part two

Some other instructions.

4 punti

Esercizio 1. Let $A = \{a,b,c\}$ and $B = \{a,c,y\}$.

(a) *(2 punti)* List (without repetition) the elements of the set $A \cup B$

Soluzione:

$$A \cup B = \{a,b,c,y\}$$

(b) *(2 punti)* List (without repetition) the elements of the set $A \cap B$

Soluzione:

$$A \cap B = \{a,c\}$$

An example of matching question.

Esercizio 2.
$a = 4$, $b = 3$, $c = 7$

3 punti

(A)	4×3	(A)	12
(B)	$4 + 3$	(B)	7
(C)	$7 - 4$	(C)	3

An example of “tabella” question.

Esercizio 3. Solve the following equations:

4 punti

Equation	Solution
$x^2 + 9x + 14 = 0$	$x = 7;\ x = 2$
$x^2 - 9x + 14 = 0$	$x = -7;\ x = -2$
$x^2 + 5x - 14 = 0$	$x = -7;\ x = 2$
$x^2 - 5x - 14 = 0$	$x = 7;\ x = -2$

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Part One

15 punti

SOME EXERCISES FROM 2 FILES

1. *(1 punto)* exercise 2b $a = 3$, $b = 2$, $c = 6$

► (a) answer 1 correct

(b) answer 2 wrong

(c) answer 5 wrong

(d) answer 3 wrong

(e) answer 4 wrong

2. *(1 punto)* exercise 2a $a = 3$, $b = 5$, $c = 6$

(a) answer 5 wrong

(b) answer 4 wrong

(c) answer 2 wrong

(d) answer 3 wrong

► (e) answer 1 correct

3. *(2 punti)* exercise 3a $a = 4$, $b = 4$, $c = 5$

(a) answer 5 wrong

(b) answer 4 wrong

(c) answer 3 wrong

► (d) answer 1 correct

(e) answer 2 wrong

Soluzione: answer 1

4. *(2 punti)* exercise 3b $a = 3$, $b = 5$, $c = 3$

(a) answer 5 wrong

(b) answer 3 wrong

(c) answer 2 wrong

(d) answer 4 wrong

► (e) answer 1 correct

Soluzione: answer 1

5. *(1 punto)* exercise 2c $a = 4$, $b = 2$, $c = 6$

► (a) answer 1 correct

(b) answer 2 wrong

(c) answer 3 wrong

(d) answer 4 wrong

(e) answer 5 wrong

6. *(1 punto)* exercise 4e $a = 2$, $b = 5$, $c = 7$

(a) answer 5 wrong

(b) answer 2 wrong

► (c) answer 1 correct

(d) answer 4 wrong

(e) answer 3 wrong

7. *(2 punti)* exercise 9c $a = 2$, $b = 3$, $c = 4$

(a) answer 4 wrong

► (b) answer 1 correct

(c) answer 2 wrong

(d) answer 5 wrong

(e) answer 3 wrong

8. *(1 punto)* exercise 1b $a = 2$

(a) answer 3 wrong

► (b) answer 1 correct

(c) answer 2 wrong

9. *(1 punto)* exercise 14a $a = 4$, $b = 3$, $c = 7$ $alphai = 5$ $alphaii = 7$

(a) answer 5 wrong

(b) answer 3 wrong

► (c) answer 1 correct

(d) answer 4 wrong

(e) answer 2 wrong

Soluzione: This is the solution $a = 4$, $b = 3$, $c = 7$

10. *(1 punto)* $a = 12$, $b = 13$, $c = 3$, $k = 5$
Evaluate with two exact decimals

$12 : 3 =$ **4**

All the problems of a file

11. *(1 punto)* exercise 11b

$$\{a,\ c,\ z\}$$

(a) answer 3 wrong

(b) answer 2 wrong

► (c) answer 1 correct

12. *(1 punto)* exercise 11a

$$\{b,\ c,\ x\}$$

► (a) answer 1 correct

(b) answer 3 wrong

(c) answer 2 wrong

Some other questions

1. *(2 punti)* exercise 13c $a = 4$, $b = 4$, $c = 6$

(a) answer 5 wrong

(b) answer 2 wrong

(c) answer 3 wrong

► (d) answer 1 correct

(e) answer 4 wrong

2. *(1 punto)* exercise 12c $a = 3$, $b = 5$, $c = 3$

► (a) answer 1 correct

(b) answer 5 wrong

(c) answer 4 wrong

(d) answer 2 wrong

(e) answer 3 wrong

Part two

Some other instructions.

4 punti

Esercizio 1. Let $A = \{a,x,y\}$ and $B = \{a,y,z\}$.

(a) *(2 punti)* List (without repetition) the elements of the set $A \cup B$

Soluzione:

$$A \cup B = \{a,x,y,z\}$$

(b) *(2 punti)* List (without repetition) the elements of the set $A \cap B$

Soluzione:

$$A \cap B = \{a,y\}$$

An example of matching question.

ESERCIZIO 2. $a = 12$, $b = 13$, $c = 2$

(C)	12×13	(A)	9
(A)	3^2	(B)	3
(B)	$36 : 12$	(C)	156
(D)	12×13	(D)	156
(E)	12×13	(E)	156

3 punti

An example of “tabella” question.

Esercizio 3. Complete the following table of derivatives:

5 punti	

Function	Derivative
$f(x) = x^2$	$f'(x) = 2x$
$f(x) = \sin x$	$f'(x) = \cos x$
$f(x) = \cos x$	$f'(x) = -\sin x$

Soluzione Versione n. 1

1. e
2. c
3. e
4. a
5. a
6. c
7. b
8. d
9. e
10. 3
11. a
12. b

1. a
2. e

Soluzione Versione n. 2

1. d
2. a
3. c
4. b
5. d
6. a
7. 3,5
8. a
9. b
10. b
11. c
12. c

1. a
2. d

Soluzione Versione n. 3

1. a
2. a
3. a
4. c
5. c
6. b
7. 3,25
8. a
9. c
10. a
11. b
12. a

1. b
2. d

Soluzione Versione n. 4

1. a
2. e
3. d
4. e
5. a
6. c
7. b
8. b
9. c
10. 4
11. c
12. a

1. d
2. a